

BILKENT UNIVERSITY

Engineering Mathematics I Midterm Exam									
Code : Math 241					Last Name:				
Acad. Year: 2015-2016					Name :			Student No.:	
Semester : Fall					Section: :			Signature:	
Date : 20.11.2015					5 QUESTIONS ON 5 PAGES TOTAL 100 POINTS				
Time : 17:40									
Duration : 120 minutes									
1	2	3	4	5					

No partial credit will be given for unsupported answers!

1. (20 pts) Let $\begin{bmatrix} a & 0 & b & | & 2 \\ a & a & 4 & | & 4 \\ 0 & a & 2 & | & b \end{bmatrix}$ be the augmented matrix of a linear system. Find for what values of a and b the system has

- (a) a unique solution
- (b) a one-parameter solution
- (c) a two-parameter solution
- (d) no solution

$$\begin{bmatrix} a & 0 & b & | & 2 \\ a & a & 4 & | & 4 \\ 0 & a & 2 & | & b \end{bmatrix} \xrightarrow{-R_1+R_2 \rightarrow R_2} \begin{bmatrix} a & 0 & b & | & 2 \\ 0 & a & 4-b & | & 2 \\ 0 & a & 2 & | & b \end{bmatrix} \xrightarrow{-R_3+R_2 \rightarrow R_3}$$

$$\begin{bmatrix} a & 0 & b & | & 2 \\ 0 & 0 & 2-b & | & 2-b \\ 0 & a & 2 & | & b \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} a & 0 & b & | & 2 \\ 0 & a & 2 & | & b \\ 0 & 0 & 2-b & | & 2-b \end{bmatrix}$$

a) To have a unique soln, we should have 3 leading variables and this is only possible when $a \neq 0$ $2-b \neq 0$ i.e. $b \neq 2$:

$$\boxed{a \neq 0 \text{ and } b \neq 2}$$

b) To have a one-parameter soln, we should have 2 leading variables, 1 free variable.

This is only possible when $\boxed{a \neq 0 \text{ and } b = 2}$

c) To have a 2 parameter soln, we should have 1 leading variable, 2 free variables. This is only possible when $a=0$ and $b=2$

d) To have no soln, we should have a bad row. This is only possible when $a=0$ and $b \neq 2$

2. (20 pts)

(a) Show that $\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$ is a basis for the subspace W of $\mathbb{R}^3$ defined as

$$W = \left\{ x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x \in \mathbb{R}^3, x_1 + x_2 + x_3 = 0 \right\}$$

(b) Find the orthogonal projection of $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ on the subspace W defined in part (a).

a) Let $S = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$

(i) S is L.I.:

Let $\alpha_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{matrix} \alpha_1 \\ \alpha_2 \end{matrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{R_3 + R_1 \rightarrow R_1}$

$$\begin{bmatrix} 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 + R_1 \rightarrow R_1} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \alpha_1 = 0, \alpha_2 = 0$$

$\Rightarrow S$ is L.I.

(ii) S spans W :

$\forall x \in W, x = \begin{bmatrix} x_1 \\ x_2 \\ -x_1 - x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \in \text{span } S$

$\Rightarrow S$ spans W .

Hence S is a basis for W .

b) Observe that $S = \left\{ \underbrace{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}}_{w_1}, \underbrace{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}}_{w_2} \right\}$ is not orthogonal

since $w_1 \cdot w_2 = 1 \neq 0$. Let's orthonormalize S by using Gram-Schmidt.

$$\hat{e}_1 = \frac{w_1}{\|w_1\|} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$\hat{e}_2 = \frac{w_2 - (w_2 \cdot \hat{e}_1) \hat{e}_1}{\|w_2 - (w_2 \cdot \hat{e}_1) \hat{e}_1\|} = \frac{\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}}{\left\| \begin{bmatrix} -1/2 \\ 1 \\ -1/2 \end{bmatrix} \right\|} = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$$

5 Hence, $\left\{ \underbrace{\begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}}_{\hat{e}_1}, \underbrace{\begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}}_{\hat{e}_2} \right\}$ is an orthonormal basis for W .

orthogonal proj. of $u = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ on W is:

$$\begin{aligned} u_{\text{proj } W} &= (u \cdot \hat{e}_1) \hat{e}_1 + (u \cdot \hat{e}_2) \hat{e}_2 \quad +2 \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} + \frac{3}{\sqrt{6}} \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \quad +3 \end{aligned}$$

3. (20 pts) Consider the matrix $A = \begin{bmatrix} 2 & 1 & 1 & 4 \\ 0 & 1 & 0 & 5 \\ 4 & 1 & 1 & 6 \end{bmatrix}$.

(a) Find a basis for the row space of A .

(b) Find a basis for the column space of A .

(c) Find a basis for the null space of A (i.e. for the solution space of $Ax = 0$).

(d) Find row rank, column rank, rank and nullity of A .

$$A \xrightarrow{-2R_1+R_3 \rightarrow R_3} \begin{bmatrix} 2 & 1 & 1 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & -1 & -1 & -2 \end{bmatrix} \xrightarrow{R_2+R_3 \rightarrow R_3} \begin{bmatrix} 2 & 1 & 1 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & -1 & 3 \end{bmatrix} \xrightarrow{\frac{1}{2}R_1 \rightarrow R_1, -1R_3 \rightarrow R_3}$$

$$\begin{bmatrix} \underline{1} & 1/2 & 1/2 & 2 \\ 0 & \underline{1} & 0 & 5 \\ 0 & 0 & \underline{1} & -3 \end{bmatrix}$$

(14)

a) A basis for the row space of A is: (+4)

$$\left\{ \begin{bmatrix} 1 & 1/2 & 1/2 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & -3 \end{bmatrix} \right\}$$

b) A basis for the column space of A is

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\} \quad (+4)$$

c) Free var: $x_4 = \alpha \Rightarrow x_3 = 3\alpha$
 $x_2 = -5\alpha$
 $x_1 = -\alpha$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \text{ is a soln of } Ax=0 \text{ iff } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ -5 \\ 3 \\ 1 \end{bmatrix}$$

$\Rightarrow$ A basis for the null space is $\left\{ \begin{bmatrix} -1 \\ -5 \\ 3 \\ 1 \end{bmatrix} \right\}$ (+4)

$$a) \text{ row rank}(A) = \text{col rank}(A) = \text{rank}(A) = \dim(\text{row}(A))$$

$$\text{nullity}(A) = \dim(\text{null}(A)) = 1 \quad \textcircled{+1} = 3 \quad \textcircled{+2}$$

4. (20 pts)

- (a) Prove that if A is $n \times n$ and $\det A \neq 0$, then $\det \text{Adj} A = (\det A)^{n-1}$
- (b) A square matrix A is called nilpotent if A^k is the zero matrix for some positive integer k .
Prove that if A is nilpotent, then A is singular.
- (c) Prove by using basic definition of the determinant that if the matrix has a zero row then its determinant is zero.

- (d) Consider the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ k & 2 & k \\ 2 & 4 & 2 \end{bmatrix}$. Find the values of k for which A is invertible.

Solution:

(5p) (a) $A^{-1} = \frac{1}{\det A} \text{Adj} A$ $\det(A^{-1}) = \det\left[\frac{1}{\det A} \text{Adj} A\right]$
1p. $\frac{1}{\det A} = \left(\frac{1}{\det A}\right)^n \det(\text{Adj} A)$
 $(\det A)^{n-1} = \det(\text{Adj} A)$

(5p) (b) $\exists k \ni A^k = 0$ $\det(A^k) = 0 \Rightarrow (\det A)^k = 0 \Rightarrow \det A = 0$
 $\Rightarrow A$ is singular.

(5p) (c) $\det A = \sum_{k=1}^n a_{jk} A_{jk}$ where a_{jk} is the k th element of the j th row and A_{jk} is the corresponding co-factor.

If the j th row is all zero then $\det A = \sum_{k=1}^n 0 \times A_{jk} = 0$
 good verbal explanation 3p.

(5p) (d) $\det \begin{bmatrix} 2 & 1 & 0 \\ k & 2 & k \\ 2 & 4 & 2 \end{bmatrix} = 2(4-4k) - (2k-2k) = 2(4-4k) = 0 \Rightarrow k=1.$

For A be invertible we need to have $\det A \neq 0$ which is possible if $k \neq 1$ ($k \in \mathbb{R} - \{1\}$)

5. Consider the matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 5 & 4 & 5 \\ 0 & 3 & 0 \end{bmatrix}$.

(a) Find A^{-1} by using elementary row operations.

(b) Find A^{-1} by using adjoint matrix method.

Solution

(10p) (a) $\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 0 & 0 \\ 5 & 4 & 5 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1/2 \rightarrow R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 5 & 4 & 5 & 0 & 1 & 0 \\ 0 & 3 & 0 & 0 & 0 & 1 \end{array} \right]$

$\xrightarrow{-5R_1 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 4 & 5 & -5/2 & 1 & 0 \\ 0 & 3 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 \leftrightarrow R_3 \\ R_2/3 \rightarrow R_2 \end{array}} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1/3 \\ 0 & 4 & 5 & -5/2 & 1 & 0 \end{array} \right]$

$\xrightarrow{-4R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 5 & -5/2 & 1 & -4/3 \end{array} \right]$

$\xrightarrow{R_3/5 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1/3 \\ 0 & 0 & 1 & -1/2 & 1/5 & -4/15 \end{array} \right] \quad A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 1/3 \\ -1/2 & 1/5 & -4/15 \end{bmatrix}$

(10p) (b) $\det A = 2(0 - 15) = -30$

$A^{-1} = \frac{1}{-30} \begin{bmatrix} -15 & 0 & 15 \\ 0 & 0 & -6 \\ 0 & -10 & 8 \end{bmatrix}^T = \frac{1}{-30} \begin{bmatrix} -15 & 0 & 0 \\ 0 & 0 & -10 \\ 15 & -6 & 8 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 0 & 1/3 \\ -1/2 & 1/5 & -4/15 \end{bmatrix}$

-3 for each mistake